MILITARY SPECIFICATION

SUPPRESSOR, SURGE, FUEL TRANSFER TRAILER

1. SCOPE

1.1 This specification covers a fuel transfer trailer surge suppressor.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-741	Steel, Carbon: Structural Shapes And Bars
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-T-5021	Tests; Aircraft And Missile Welding Operators' Qualifications
MIL-J-5624	Jet Fuel, Grades JP-4 And JP-5
MIL-F-26887	Fuel Servicing Unit, Aircraft, Trailer Mounted, Type MH-2A (Trailer, Fuel Transfer, Filter-Meter-Hose, Type MH-2A)
MIL-D-70327	Drawings, Engineering And Associated Lists

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of
MIL-STD- 808	Finishes, Protective And Codes For Finishing Schemes For Ground And Ground Support Equipment
MIL-STD-810	Environmental Test Methods For Aerospace And Ground Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS28889	Valve, Air, High-Pressure Charging

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with

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specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance With MIL-STD-143.

3.3 Materials

3.3.1 Bellows or piston assembly, housing orifice, and inlet connection. The bellows assembly, housing, orifice, and inlet connection shall be fabricated of 301, 302. or 303 stainless-steel. The piston assembly shall be fabricated of 301, 302, or 303 stainless-steel or 2024, 5052, or 6061 aluminum alloy suitably treated for the application.

3.3.2 Removable caps. The removable caps specified in 3.4.2.4 and 3.4.2.5 shall be of corosion-resistant steel and attached to the housing with nonferrous metal chains.

3.4 Design and construction. The suppressor shall incorporate a diaphragm-type bellows or piston assembly within a pressurized housing to absorb surge pressures encountered in pipelines. Nitrogen shall be used for pressurization of the housing. The bellows or piston assembly and the housing shall be designed and constructed to permit the required expansion or compression of the bellows or movement of the piston within the housing.

3.4.1 Bellows or piston assembly, The bellows shall be of the multiple construction, diaphragm type with heli-arc welds at the inner and outer peripheries of the diaphragms. The bellows or piston assembly shall be capable of extending or compressing sufficiently to accommodate a minimum of 465 cubic inches of fluid displaced from the associated piping system.

3.4.1.1 Life expectancy. The bellows or piston assembly shall be designed for a minimum life of 100,000 cycles at full travel.

3.4.1.2 Hydrostatic pressure. The bellows or piston assembly shall be capable of withstanding a hydrostatic pressure of 150 psig against a gas pressure of 52 psig within the suppressor housing. The housing shall be capable of withstanding a hydrostatic pressure of 300 psig.

3.4.1.2.1 Nitrogen gas precharging. The suppressor shall be precharged with nitrogen gas to 52 ±2 psig at 70°F.

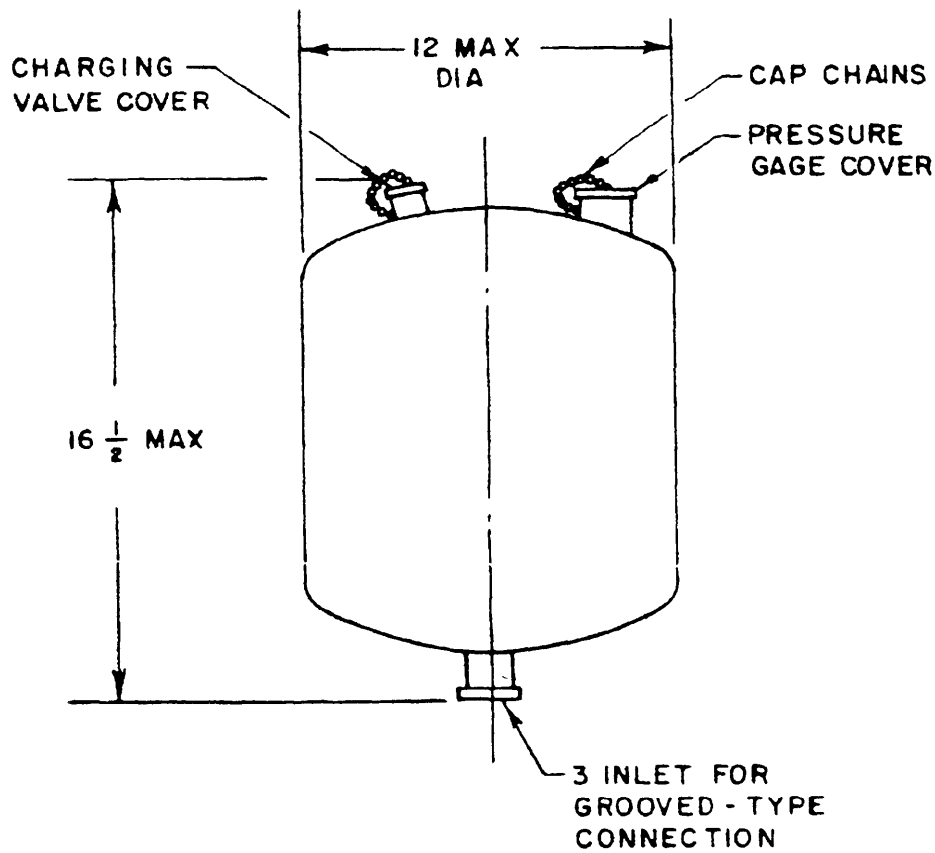
3.4.2 Housing. The housing shall be designed to meet the requirements specified herein.

3.4.2.1 Configuration. The configuration of the housing shall be as shown on figure 1.

3.4.2.2 Gas capacity. The minimum gas capacity of the housing, with the bellows assembly or piston completely retracted, shall be 738 cubic inches.

3.4.2.3 Flow-limiting orifice. A flow-limiting orifice shall be

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES : $\pm \frac{1}{32}$

FIGURE 1. Surge Suppressor Housing

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incorporated at the inlet to the housing. The orifice opening shall be 1.309 ±0.002 inch in diameter.

3.4.2.4 Pressurizing gas connection. A valve conforming to MS28889 shall be installed in the upper head of the housing. The valve shall be protected by a cover and removable cap. The protective cover shall be so designed that a standard tire chuck may be used in applying gas pressure to the valve.

3.4.2.5 Pressure gage. A pressure gage of good commercial quality shall be installed in the upper head of the housing, It shall have a dial with a 1-1/2-inch minimum diameter and a range of from 0 to 160 psi. The pressure gage shall be protected by a cover and removable cap.

3.4.2.6 Inlet fluid connection. The inlet fluid connection shall be 3-inch nominal size, grooved without a backup shoulder, in accordance with tubing end fittings as specified in MIL-F-26887.

3.4.2.7 Handtools. The housing shall be so constructed that the bellows or piston can be readily replaced by using activities through the use of common handtools.

3.4.3 Mounting. The mounting shall be as shown on figure 2 and shall be constructed of 1-1/2-by-1-1/2-by-3/16-inch structural steel angles and triangular base plates 1-3/4 inches on the two equal sides and 3/16 inch thick with both the plate and angle material conforming to QQ-S-741. For suppressors furnished on new fueling vehicles, the legs shall be as shown on figure 2. The base plates shall be welded flush with the outside edge of the angles. Unless otherwise specified (see 6.2), base plate's shall be delivered as separate items for surge suppressors furnished as replacement items, These base plates shall be welded on the legs by the field activity after the proper length of the legs has been established.

3.4.4 Certification of welders. Welders employed in the fabrication of the suppressor shall be certified in accordance with MIL-T-5021.

3.5 Performance

3.5.1 Environmental conditions. The suppressor shall be capable of satisfactory operation when subjected to the following environmental conditions:

a. Ambient temperatures ranging from +125° to -65°F

b. A relative humidity of 95 ±5 percent at 160° ±2°F

c. Airborne sand and dust particles encountered in normal and desert operations

d. Atmosphere containing salt-laden moisture

e. Fungus encountered in tropical climates.

3.5.1.1 Storage temperatures. The suppressor shall be capable of withstanding storage in temperatures ranging from +160° to -80°F.

3.5.2 Surge. The suppressor shall be capable of withstanding a maximum surge of from 0 to 120 psig in 0.5 second, or less.

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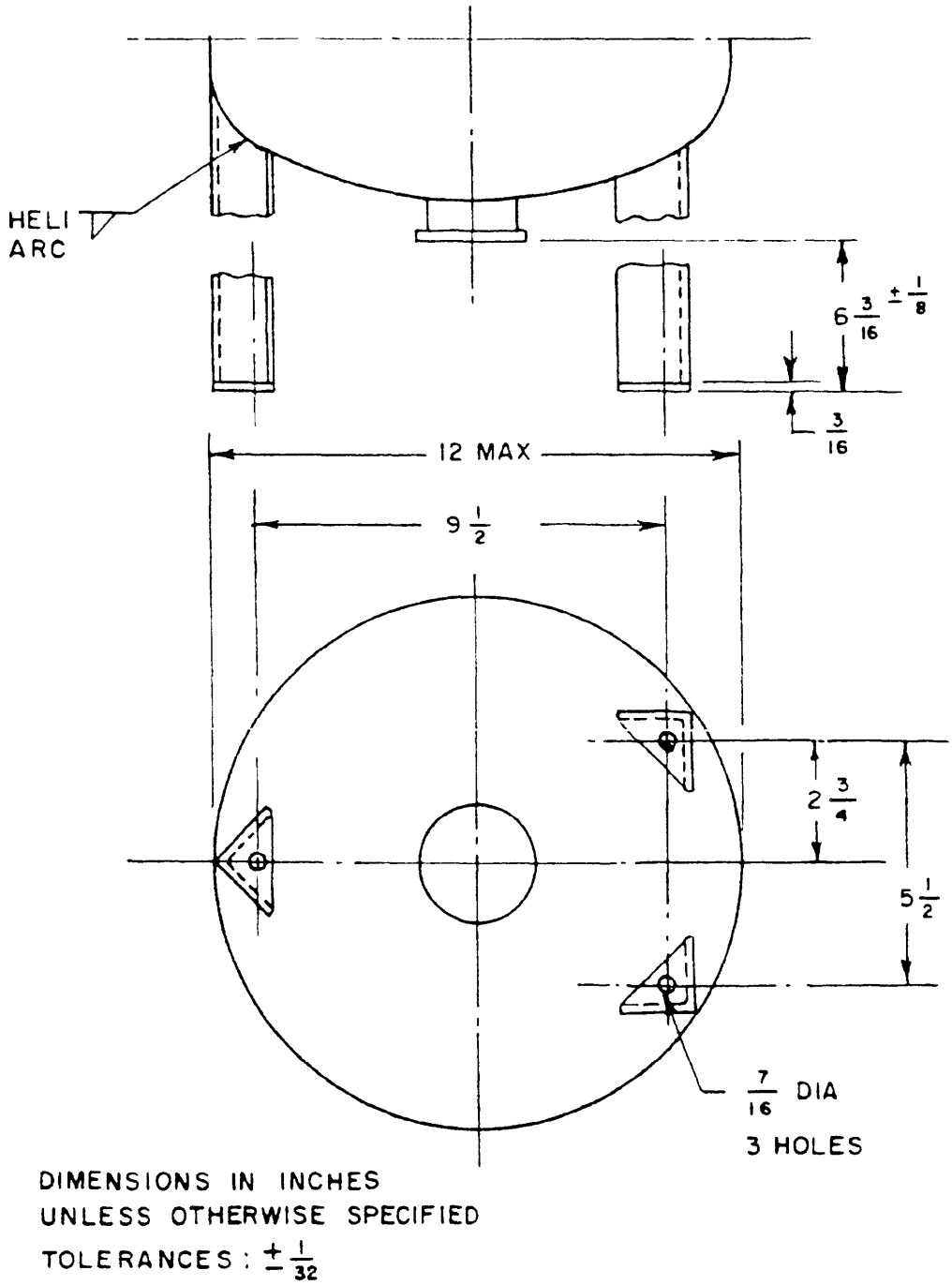


FIGURE 2. Surge Suppressor Mounting

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3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's a part number shall be functionally and dimensionally Interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer' s part numbers and changes thereto.

3.7 Weight. The weight of the suppressor shall not exceed 70 pounds.

3.8 Finish. The suppressor shall be finished in accordance with MIL-STD-808, type I, color code DG.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. Workmanship shall be of the highest grade and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the suppressor shall be classified as folllows:

a. Preproduction testing. .See 4.4

b. Acceptance tests . . . .See 4.5.

4.3 Test condition

4. 3.1 Test fluid. The test fluid shall be grade JP-4 or JP-5 fuel conforming to MIL-F-5624. The specific gravity and density of the test fluid at standard conditions shall be included in the test report.

4.4 Preproduction testing (see 6.2)

4.4.1 Preproduction test sample. One suppressor shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Preproduction test report. After the preproduction tests are completed, three copies of a test report prepared in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.4.2.1 The test report shall include copies of the drawings used in the manufacture of the item, 8-by-10 black and white photographs showing a minimum of three views of the item and the test setup, and copies of the recording chart used during the cycle test.

4.4.2.2 Maintainability information. The information obtained when demonstrated in accordance with 4.6.6 shall be submitted as an attachment accompanying the preproduction test report (same quantity and quality of copies) or shall be included as part of the report.

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4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each suppressor shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Hydrostatic pressure test . . . . See 4.6.2.

4.6 Test methods

4.6.1 Examination of product. The suppressor shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Hydrostatic pressure test. The suppressor housing shall be hydrostatically tested to 300 psig. The bellows assembly shall be hydrostatically tested with an internal pressure of 150 psig against a precharging gas pressure of 52 ±2 psig. The housing and bellows assembly shall not leak at the pressures specified.

4.6.3 Environmental tests. The suppressor shall be subjected to the following tests conducted in accordance with the specified procedures of M1L-STD-810. An engineering evaluation concerning the ability of the equipment to resist the exposure specified in 4.6.3.2 through 4.6.3.5 shall be considered as meeting the tests specified herein.

4.6.3.1 Surge cycling tests

4.6.3.1.1 Low temperature test. The suppressor shall be subjected to low temperature in accordance with method 502, procedure I. At the conclusion of the exposure period and while at the specified temperature, the suppressor shall be subjected to a minimum of 20 cycles of fluid pressure alternating between 0 and 120 psig applied to the inlet port. The pressure rise from 0 to 120 psig at the inlet port shall be accomplished in 0.5 second, or less. At the conclusion of the test, the suppressor shall be visually examined for evidence of failure or deterioration. There shall be no internal nor external leakage.

4.6.3.1.2 High temperature test. The suppressor shall be subjected to high temperature in accordance with method 501, procedure I, except as specified herein. At the conclusion of the exposure period, the temperature of the chamber shall be lowered to 125°F. The suppressor shall then be subjected to a minimum of 20 cycles of fluid pressure alternating between 0 and 120 psig applied to the inlet port. The pressure rise from 0 to 120 psig at the inlet port shall be accomplished in 0.5 second, or less. At the conclusion of the test, the suppressor shall be visually examined for evidence of failure or deterioration. There shall be no internal nor external leakage as a result of this test.

4.6.3.2 Humidity test. The suppressor shall be subjected to humidity in accordance with method 507 procedure I. At the conclusion of the exposure period and while at room temperature, the suppressor shall

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be subjected to the surge cycling specified in 4.6.3.1.1.

4.6.3.3 Sand and dust test. The suppressor shall be subjected to sand and dust in accordance with method 510, procedure I. At the conclusion of the test, the suppressor shall be subjected to the surge cycling specified in 4.6.3.1.1.

4.6.3.4 Salt-fog test. The suppressor shall be subjected to salt fog in accordance with method 509. At the conclusion of the test, the suppressor shall be subjected to the surge cycling specified in 4.6.3.1.1.

4.6.3.5 Fungus test. The suppressor shall be subjected to fungus in accordance with method 508, procedure I.

4.6.4 Cycling test. The suppressor shall be subjected to a minimum of 100,000 cycles of fluid pressure alternating between 0 and 120 psig applied to the inlet port. The pressure rise from 0 to 120 psig at the inlet port shall be accomplished in 0.5 second, or less. At the conclusion of the tests, the suppressor shall be inspected for damage and loss of nitrogen gas pressure.

4.6.4.1 Gas recharging test. upon completion of the cycling test, the nitrogen pressure in the housing shall be checked. The nitrogen pressure shall be 52 ±2 psig at 70°F.

4.6.5 Surge control demonstration. Surge arrestors of designs not previously approved or supplied as original equipment for the hose trailers or trucks shall be installed on a Type MH-2A filter-meter-hose trailer and subjected to the surge control test of MIL-F-26887. With the arrestors Installed, all requirements for the surge control shall be met.

4.6.6 Maintainability demonstration. The bellows or piston snail be removed and replaced and the following information recorded:

a. Actual clocktime to do the task

b. The tools required

c. Any obstruction or projections that hindered the task or that presented potential damage or hazard to bellows or piston or to the personnel doing the task.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with section 5 or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C (see 6.2).

5.1.1 Level A. The surge suppressor shall be preserved as specified in method III of MIL-P-116.

5.1.2 Level C. The surge suppressor shall be packaged for mechanical and physical protection from the contractor's facility to designated designations.

5.2 Packing. Packing shall be level A, B, or C (see 6.2).

5.2.1 Level A. Suppressors, preserved as specified in 5.1, shall be packed one each in a container

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conforming to PIP-B-636, type SF, class weather-resistant, grade VIs. Closure and strapping shall be in accordance with the container specification and appendix thereto.

5.2.2 Level B. Suppressors, preserved as specified in 5.1, shall be packed one each in a container conforming to PPP-B-636, type SF, class domestic, grade 275. Closure and strapping shall be in accordance with the container specification and appendix thereto.

5.2.3 Level C. Suppressors, preserved as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Blocking, bracing, cushioning, and testing. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Packs shall be tested in accordance with the applicable paragraphs of appendix A of MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

SUPPRESSOR, SURGE, FUEL
TRANSFER TRAILER

6. NOTES

6.1 Intended use. The suppressor covered by this specification is intended for use as an integral part of a gasoline or jet fuel dispensing system, for use on a Type MH-2 fuel transfer trailer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether base plates are to be furnished welded to legs for replacement items (see 3.4.3)

c. Location and conditions for preproduction testing (see 4.4)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
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COMMANDER
SYSTEMS ENGINEERING GROUP
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WRIGHT-PATTERSON AFB, OHIO 45433

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 85 1426 REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE. OBSOLETE C-0070

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